

3.10.3. Computation of transcript TT

Mapping (Version 1.0)

Crypto_Transcript(PBKDFParamRequest, PBKDFParamResponse, pA, pB) :=
byte[] TT

Crypto_Transcript() SHALL compute TT as specified in SPAKE2+ with:

byte ContextPrefixValue [26] = {
0x43, 0x48, 0x49, 0x50, 0x20, 0x50, 0x41, 0x4b, 0x45, 0x20, 0x56, 0x31, 0x20, 0x43,
0x6f, 0x6d,
0x6d, 0x69, 0x73, 0x73, 0x69, 0x6f, 0x6e, 0x69, 0x6e, 0x67
} // "CHIP PAKE V1 Commissioning" - The usage of CHIP here is intentional and due to
implementation in the SDK before the name change, should not be renamed to Matter.

Context := Crypto_Hash(ContextPrefixValue || PBKDFParamRequest || PBKDFParamResponse)

TT :=
lengthInBytes(Context) || Context ||
0x0000000000000000 || 0x0000000000000000 ||
lengthInBytes(M) || M ||
lengthInBytes(N) || N ||
lengthInBytes(pA) || pA ||
lengthInBytes(pB) || pB ||
lengthInBytes(Z) || Z ||
lengthInBytes(V) || V ||
lengthInBytes(w0) || w0 ||

Z and V SHALL be computed from pA and pB as specified in SPAKE2+.

Note the two 0x0000000000000000 null-lengths indicate that no identities are present and each null-lengths is 8 bytes wide since it is specified by the SPAKE2+ specification that lengths are eight-byte little-endian numbers. The SPAKE2+ specification indicates that we **must** include these length fields.

Note in case PBKDFParamRequest and PBKDFParamResponse messages are not exchanged, they SHALL be replaced by empty strings in the Context computation.

3.10.4. Computation of cA, cB and Ke